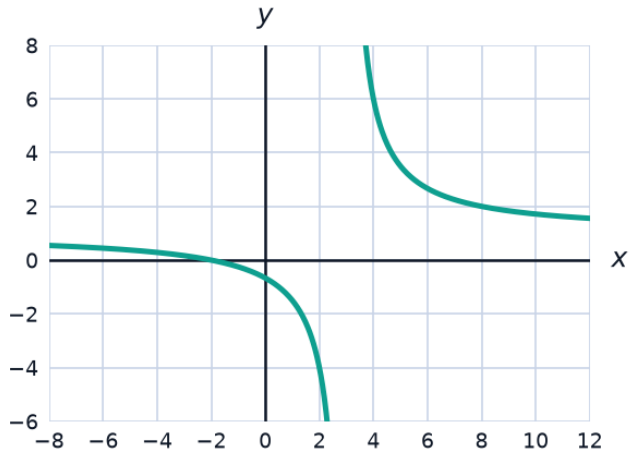
Graphing Rational Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Reading graphs

The graph of $f(x) = \frac{x+2}{x-3}$ is shown. From the graph and the formula, state the asymptotes and intercepts.



2 [Foundational] • Target: Reading graphs

The function $g(x) = \frac{1}{x-2} + 3$ is a transformation of $y = \frac{1}{x}$.



a Describe the transformation.

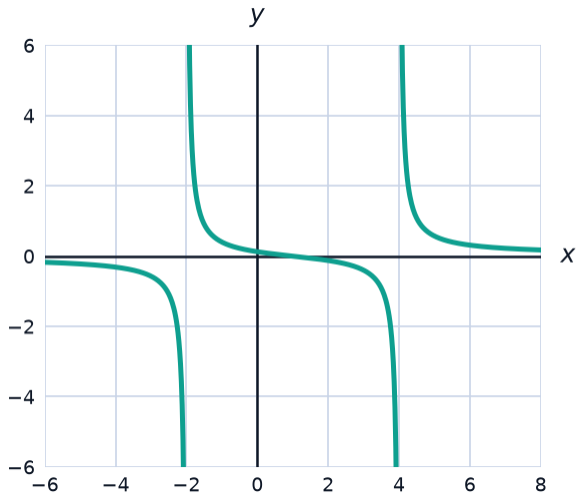
b State the asymptotes of g.

c State the domain and range.

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3 [Foundational] • Target: Sign analysis

Let $f(x) = \frac{x-1}{(x+2)(x-4)}$. Using a sign chart with critical values $x = -2, 1, 4$, determine the intervals on which $f(x) > 0$.



4 [Foundational] • Target: Sign analysis

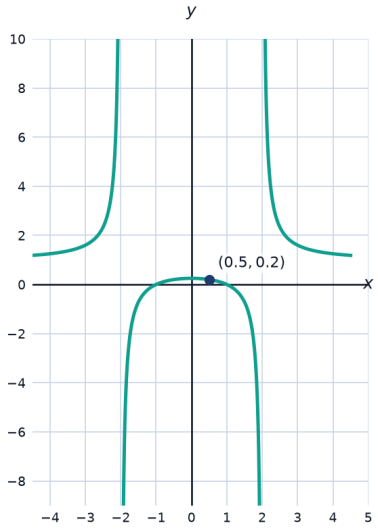
Solve the rational inequality $\frac{x+3}{x-2} \leq 0$.

5 [Foundational] • Target: Sign analysis

Sketch-planning: for $h(x) = \frac{2x}{x^2-1}$, state the asymptotes and the symmetry of the graph.

6 [Foundational] • Target: Sketching a complete graph

For $f(x) = \frac{x^2-1}{x^2-4}$, find all intercepts and asymptotes, then describe the graph's behaviour near each vertical asymptote.



7 [Foundational] • Target: Sketching a complete graph

Sketch-planning: the function $g(x) = \frac{3}{x-1} - 2$ has a vertical asymptote at $x = 1$ and horizontal asymptote $y = -2$. Find its x- and y-intercepts.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Building a rational function from a description**

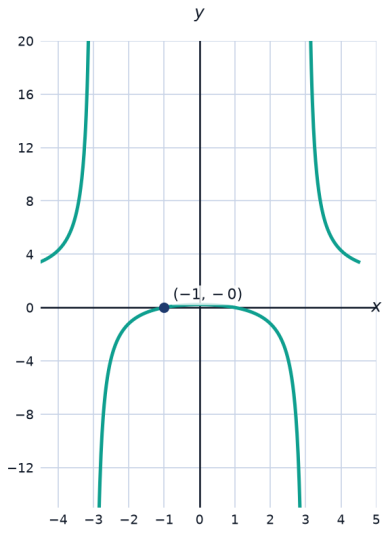
Construct a rational function with vertical asymptote $x = 2$, horizontal asymptote $y = 1$, and a hole at $x = -3$.

9 **[Intermediate]** • **Target: Graphing with a slant asymptote**

For $f(x) = \frac{x^2 - 4}{x - 1}$, find the slant asymptote and the vertical asymptote, and use them together with intercepts to describe the overall shape of the graph.

10 **[Intermediate]** • **Target: Full graph analysis: putting it all together**

For $f(x) = \frac{2x^2 - 2}{x^2 - 9}$, find all intercepts, all asymptotes, and describe the symmetry of the graph (even, odd, or neither).



11 **[Intermediate]** • **Target: Graphing near a hole**

For $f(x) = \frac{x^2 - 9}{x - 3}$, describe how the graph looks immediately around $x = 3$, and find the coordinates of the hole.

12 **[Intermediate]** • **Target: Graphing near a hole**

A rectangular garden's cost per square meter to fence is modeled by $C(x) = \frac{200}{x} + 8x$, where x is the garden's width in meters. Find the vertical asymptote and horizontal/slant asymptote of this function, and interpret what the vertical asymptote means physically (why $x = 0$ isn't a realistic input).

13 **[Intermediate]** • **Target: Graphing near a hole**

State the domain of $f(x) = \frac{\sqrt{x}}{x - 4}$.

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14 [Intermediate] • Target: Building a rational function from a description

Level 2 · MediumThe function $f(x) = x^3 - 3x^2 - 9x + 2$ has a local maximum and a local minimum. Determine the intervals on which f is concave up and concave down, and find the coordinates of the inflection point.

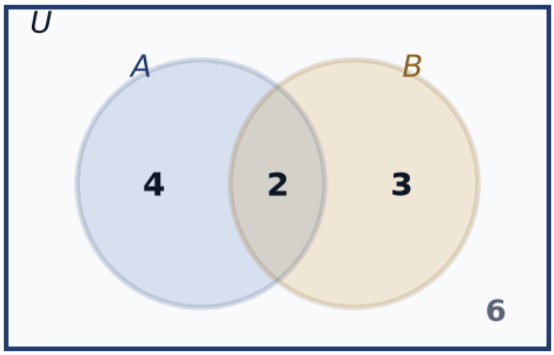
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Graphing with a slant asymptote • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

The function $f(x) = \frac{4x-7}{2x+8}$ has horizontal asymptote $y = H$ and vertical asymptote $x = V$. Determine the value of $H + V$.

16 [Challenging] • Target: Full graph analysis: putting it all together • Cross-Lesson: 4.01 Circles and Parabolas

Solve the rational inequality $\frac{x+4}{x-2} \leq 3$ and express the solution set in interval notation.



17 [Challenging] • Target: A compound rational inequality • Cross-Lesson: 4.02 Ellipses

Level 3 · AdvancedExperimental laboratory data pairs (x, y) are transformed by computing $Y = \ln(y)$. A linear regression performed on (x, Y) yields the best-fit line $Y = -0.42x + 2.85$. Write the corresponding non-linear model for y as an exponential function of x in the form $y = ab^x$, giving a and b to three decimal places. Then predict the value of y when $x = 5$.

18 [Challenging] • Target: Graphing near a hole • Cross-Lesson: 4.03 Hyperbolas

Find all real and complex roots of the polynomial $P(x) = 2x^3 + 3x^2 - 8x + 3$.

19 [Challenging] • Target: Reading graphs • Cross-Lesson: 5.01 Vectors: Components and Operations

Solve the radical equation $\sqrt{2x + 7} - x = 2$ for all real solutions and confirm extraneous roots.

20 [Challenging] • Target: Sign analysis • Cross-Lesson: 5.02 The Dot Product

Find the set of values of x satisfying the inequality $2x^2 - 7x + 3 \leq 0$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Reading graphs

Which function has a slant asymptote?

a. $f(x) = \frac{1}{x^2}$

b. $f(x) = \frac{x}{x^2+1}$

c. $f(x) = \frac{x^2+1}{x}$

d. $f(x) = \frac{x+1}{x+2}$

22 [Foundational] • Target: Sign analysis

Which function has no vertical asymptote?

a. $f(x) = \frac{x}{x-1}$

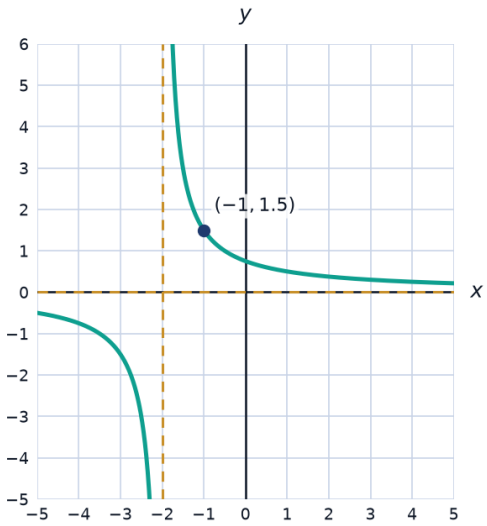
b. $f(x) = \frac{1}{x^2}$

c. $f(x) = \frac{x^2+1}{x^2+1} = 1$

d. $f(x) = \frac{1}{x}$

23 [Foundational] • Target: Sketching a complete graph

Find all rational functions with vertical asymptotes at $x = \pm 1$ and horizontal asymptote $y = 0$.



a. $\frac{x}{x^2-1}$ only

b. $\frac{1}{x-1}$ only

c. Any polynomial

d. Functions of the form $\frac{k}{x^2-1}$

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24 [Foundational] • Target: More rational inequalities

Which is the graph of an even function?

a. Symmetric about the y-axis

b. Symmetric about the origin

c. Symmetric about the x-axis

d. Has no symmetry

25 [Foundational] • Target: Building a rational function from a description

What is the maximum value of the function $f(x) = \sqrt{3}\sin(2x) - \cos(2x) - 2\sin x + 2\sqrt{3}\cos x$ for all real numbers x?

a. 3

b. 2

c. 6

d. 1

26 [Foundational] • Target: Graphing with a slant asymptote

A function f is defined for all real numbers x such that $x \neq 0$ and $x \neq 1$. If $f(x) + 2f\left(\frac{1}{1-x}\right) = 3x$ for all permissible values of x, what is the value of $f(3)$?

a. $\frac{20}{9}$

b. $\frac{7}{3}$

c. $\frac{16}{9}$

d. 3

27 [Foundational] • Target: Full graph analysis: putting it all together

The function f is defined for all real numbers $x \neq -2$ by $f(x) = \frac{4x-1}{x+2}$ What is the value of $f^{-1}(3)$?

a. -5

b. 3

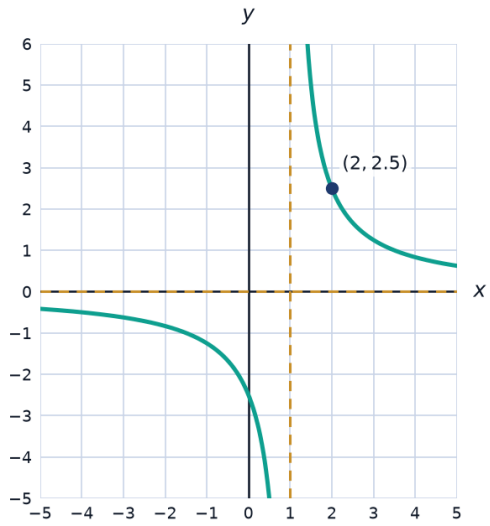
c. 5

d. 7

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28 [Foundational] • Target: A compound rational inequality

Which function is decreasing on all of $\mathbb{R}$?



a. $y = e^x$

b. $y = x^2$

c. $y = e^{-x}$

d. $y = |x|$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Graphing near a hole

Find all solutions of $\frac{x^2+2x}{x^2-1} = 1 + \frac{3}{x-1}$.

a. $x = 1$

b. No solution

c. $x = -4$

d. $x = 4$

30 [Intermediate] • Target: Reading graphs

Vertical asymptote of $\frac{x+1}{x-4}$:

a. $x = 1$

b. $x = 4$

c. $x = -1$

d. $x = -4$

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31 [Intermediate] • Target: Sign analysis

Identify: $\frac{p(x)}{q(x)}$ is called a _____ function.

a. Irrational

b. Rational

c. Polynomial

d. Transcendental

32 [Intermediate] • Target: Sketching a complete graph

Describe the graph of $f(x) = x^3(x-2)^2$ near $x = 0$ and $x = 2$.

a. Crosses at both

b. Crosses at $x = 0$; touches at $x = 2$

c. Touches at $x = 0$; crosses at $x = 2$

d. Touches at both

33 [Intermediate] • Target: More rational inequalities

A function is defined by $f(x) = 2x - 4$ for all x near $x = -4$ (including at $x = -4$). What is $\lim_{x \to -4} f(x)$?

a. -15

b. -12

c. -14

d. -10

34 [Intermediate] • Target: Building a rational function from a description

A rational function $\frac{P(x)}{Q(x)}$ where $\deg P > \deg Q$ may have:

a. No asymptote

b. A horizontal asymptote $y = 1$

c. An oblique (slant) asymptote

d. A horizontal asymptote $y = 0$

35 [Intermediate] • Target: Graphing with a slant asymptote

Find all rational zeros of $P(x) = x^3 - 7x + 6$.

a. $x = 1, 2, -3$

b. $x = 1, -2, 3$

c. $x = -1, -2, 3$

d. $x = -1, 2, 3$

36 [Intermediate] • Target: Full graph analysis: putting it all together

Find the oblique asymptote of $f(x) = \frac{x^2+1}{x}$.

a. $y = 0$

b. $y = x + 1$

c. $y = x$

d. $y = 1$

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37 [Intermediate] • Target: A compound rational inequality

The polynomial $p(x) = x^4 - 5x^2 + 4$. Which of the following are all its real zeros?

a. $x = \pm 1, \pm 2$

b. $x = \pm 2$

c. $x = 1, 2$

d. $x = \pm 1$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Graphing near a hole • Cross-Lesson: 4.03 Hyperbolas

Find all x-intercepts of $f(x) = \frac{x^2 - 4}{x + 3}$.

a. $x = \pm 2$

b. $x = -3$

c. $x = 4$

d. $x = 2$

39 [Challenging] • Target: Reading graphs • Cross-Lesson: 5.01 Vectors: Components and Operations

If $f(x) = \frac{P(x)}{Q(x)}$ with $\deg P = \deg Q$, what is the horizontal asymptote?

a. No horizontal asymptote

b. $y = \frac{\text{leading coeff of } P}{\text{leading coeff of } Q}$

c. $y = 0$

d. $y = 1$

40 [Challenging] • Target: Sign analysis • Cross-Lesson: 5.02 The Dot Product

Find all values excluded from the domain of $f(x) = \frac{x^2 - 1}{x^3 - x}$.

a. $x = 0, \pm 1$

b. $x = 0$

c. $x = \pm 1$

d. $x = 1$

41 [Challenging] • Target: Sketching a complete graph • Cross-Lesson: 5.03 Parametric Equations

A function f is measurable and satisfies $f(x + y) = f(x) + f(y)$ a.e. Then:

a. f may be discontinuous

b. $f(x) = cx$ a.e.

c. $f(x) = 0$ a.e.

d. $f(x) = cx^2$

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42 [Challenging] • Target: More rational inequalities • Cross-Lesson: 6.01 Polar Coordinates and Conversions

The graph of $y = \frac{1}{x-2}$ is the graph of $y = \frac{1}{x}$ shifted:

a. Right 2 units

b. Down 2 units

c. Up 2 units

d. Left 2 units

43 [Challenging] • Target: Building a rational function from a description • Cross-Lesson: 6.02 Polar Equations and Graphs

Determine all values of x excluded from the domain of $\frac{x^3+1}{x^2+x}$.

a. $x = -1$ only

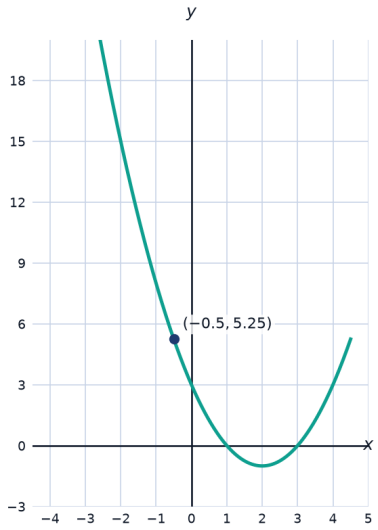
b. All real x

c. $x = 0, -1$

d. $x = 0, 1$

44 [Challenging] • Target: Graphing with a slant asymptote • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

A quadratic function is given by $f(x) = x^2 - 4x + 3$. The graph of this function is a parabola. Which of the following points lies on the graph of $f(x)$?



a. $(1, 0)$

b. $(0, 1)$

c. $(2, 3)$

d. $(4, 4)$

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45 [Challenging] • Target: Full graph analysis: putting it all together • Cross-Lesson: 7.02 Logarithms and Their Properties

The rational function f is defined by $f(x) = \frac{2x^4 - 3x^3 + kx^2 - 8x + 3}{(x-2)(x^2+1)}$, where k is a real constant. Given that the series expansion of $f(x)$ in ascending powers of x, valid for sufficiently small $|x|$, has no term in x^2, find the value of k.

a. $\frac{25}{4}$

b. $\frac{17}{4}$

c. $-\frac{25}{4}$

d. $\frac{115}{12}$

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